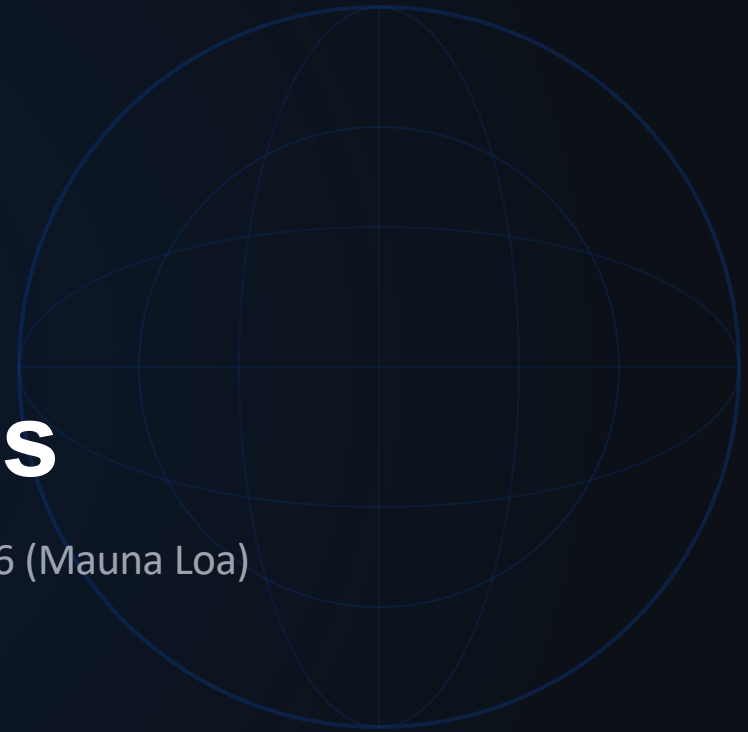


Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1750–2024 (emissions) • Atmospheric context through Feb 2026 (Mauna Loa)

Data: Global Carbon Budget 2025 (Our World in Data) • NASA Climate Vital Signs



Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

Latest Mauna Loa reading

429 ppm

February 2026 • NOAA Mauna Loa Observatory

Baseline: ~280 ppm



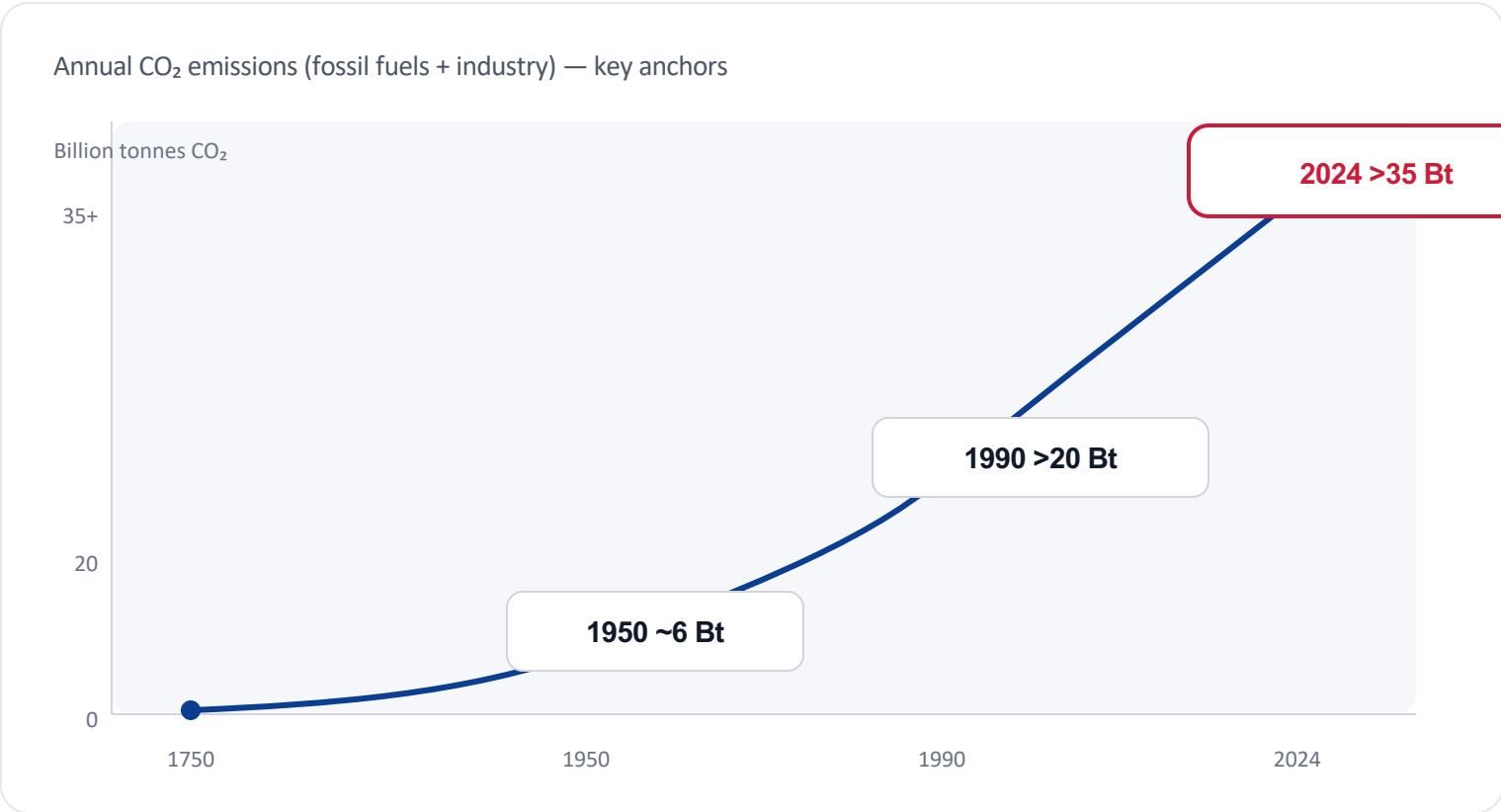
Why it matters

- ~50% rise from pre-industrial levels (~280 ppm → 429 ppm)
- ≥1.5× the natural post-ice-age increase (but compressed into ~2 centuries)
- Linked primarily to burning coal, oil, and natural gas

Acceleration since measurement began



Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes



Inflection points

- Slow growth pre-1950, then rapid acceleration mid-20th century
- 1950 → 1990: ~6 Bt → >20 Bt (near quadrupling)
- Today: >35 Bt; growth slowed, but not yet peaked

Figure reference

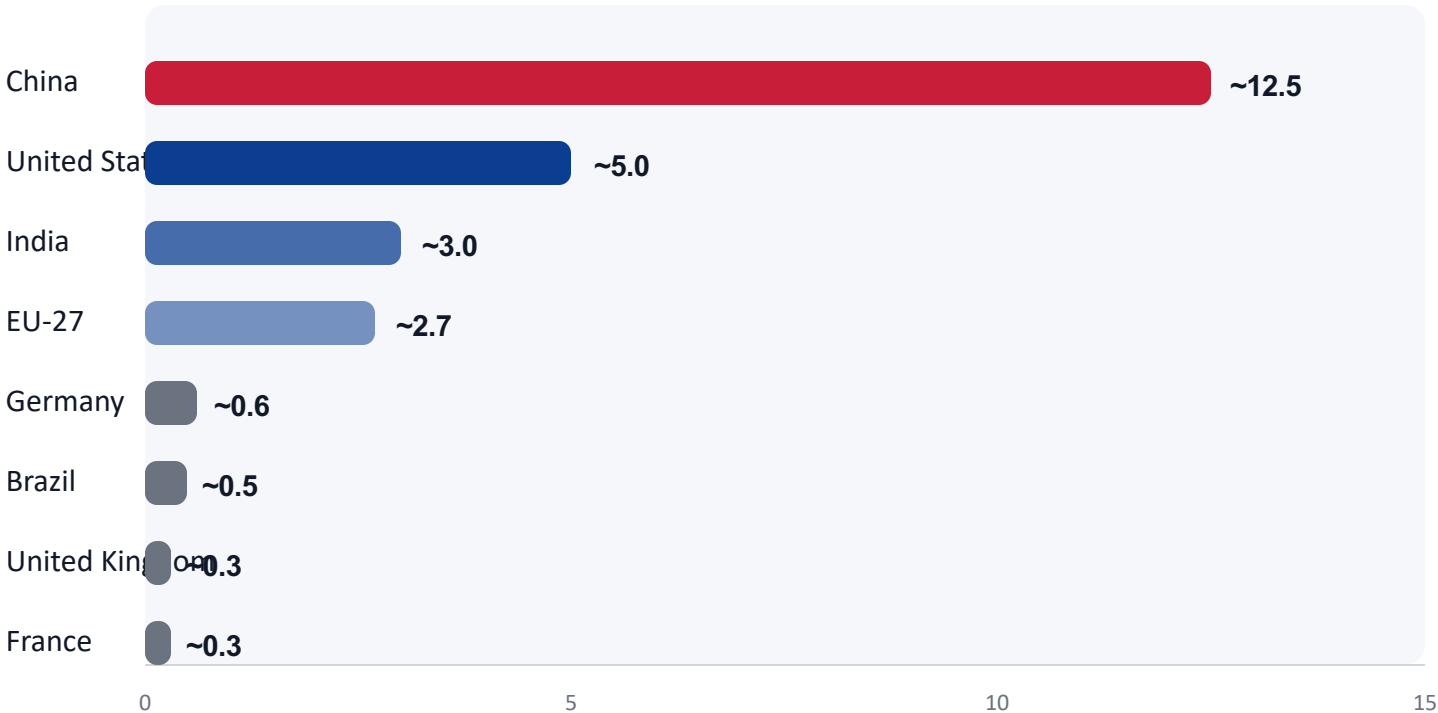
Task spec references **figure_2.jpg**, but it is not present in provided sources.

This slide uses source-brief anchor values only.

China emits ~12.5 Bt (~27%) — more than double the US (~5.0 Bt), with India (~3.0 Bt) rising rapidly

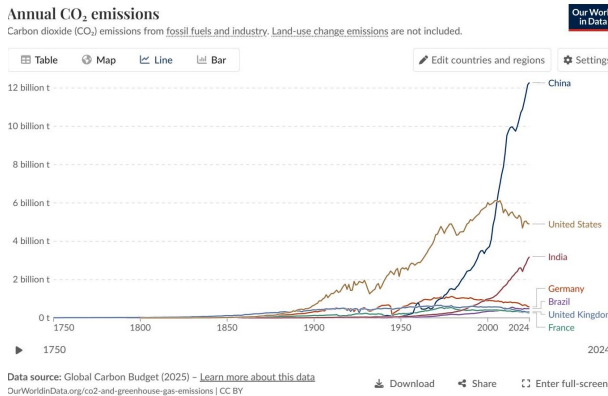
China Now Emits Over 12 Billion Tonnes — More Than Double the United States

Approx. annual CO₂ emissions (billion tonnes, fossil + industry)



Reference

Annual CO₂ emissions by country (figure_3.jpg)



Notes

- US peaked near ~6 Bt around 2005 and declined to ~5 Bt
- China is ~27% of global total; India ~8% and rising

Regional responsibility has shifted: 1900 Europe+US >90% → 2024 Asia ~50%, while US+Europe fall below one-third

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

1900 (share of global emissions)

Europe + US >90% (rest of world <10%)



- Europe + United States (dominant share)
- Rest of world

2024 (share of global emissions)

Asia ~50%; US+Europe <33%; Africa + S. America each 3–4%



- Asia (~50%)
- US + Europe (< one-third)
- Africa (3–4%)
- South America (3–4%)
- Other / remainder (~9%)



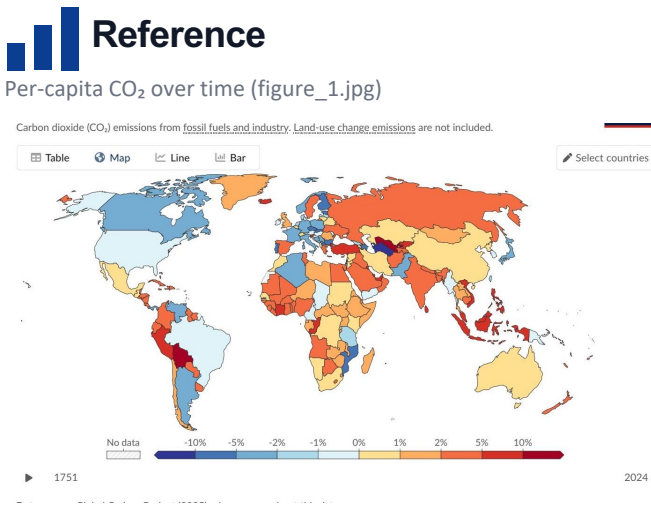
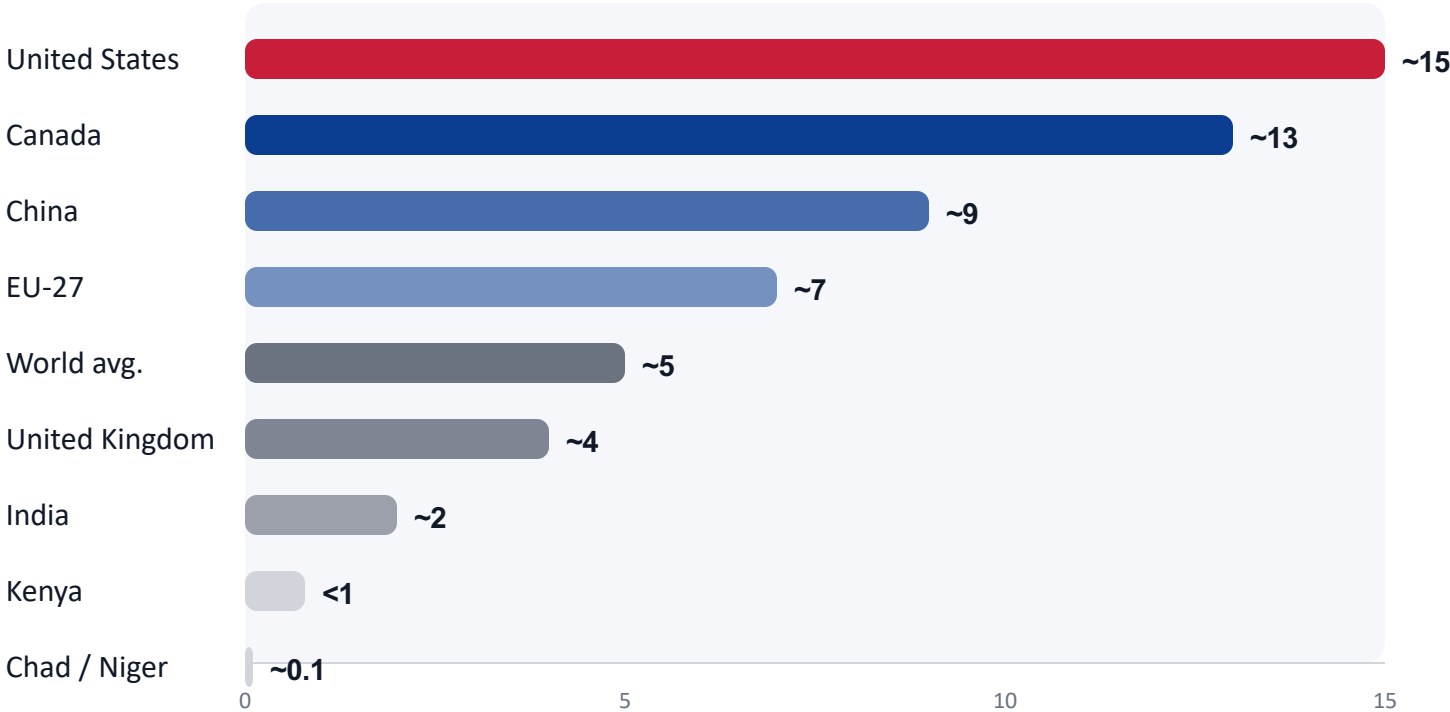
Interpretation

- Emissions leadership has shifted toward Asia as industrial output and energy demand expanded
- Africa and South America remain small shares (3–4% each), relevant for equity and growth pathways

Per-capita inequality remains extreme: ~15 t/person (US) vs ~0.1 t/person (Chad/Niger) — ~150× gap

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

Annual CO₂ per person (tonnes/year) — selected countries

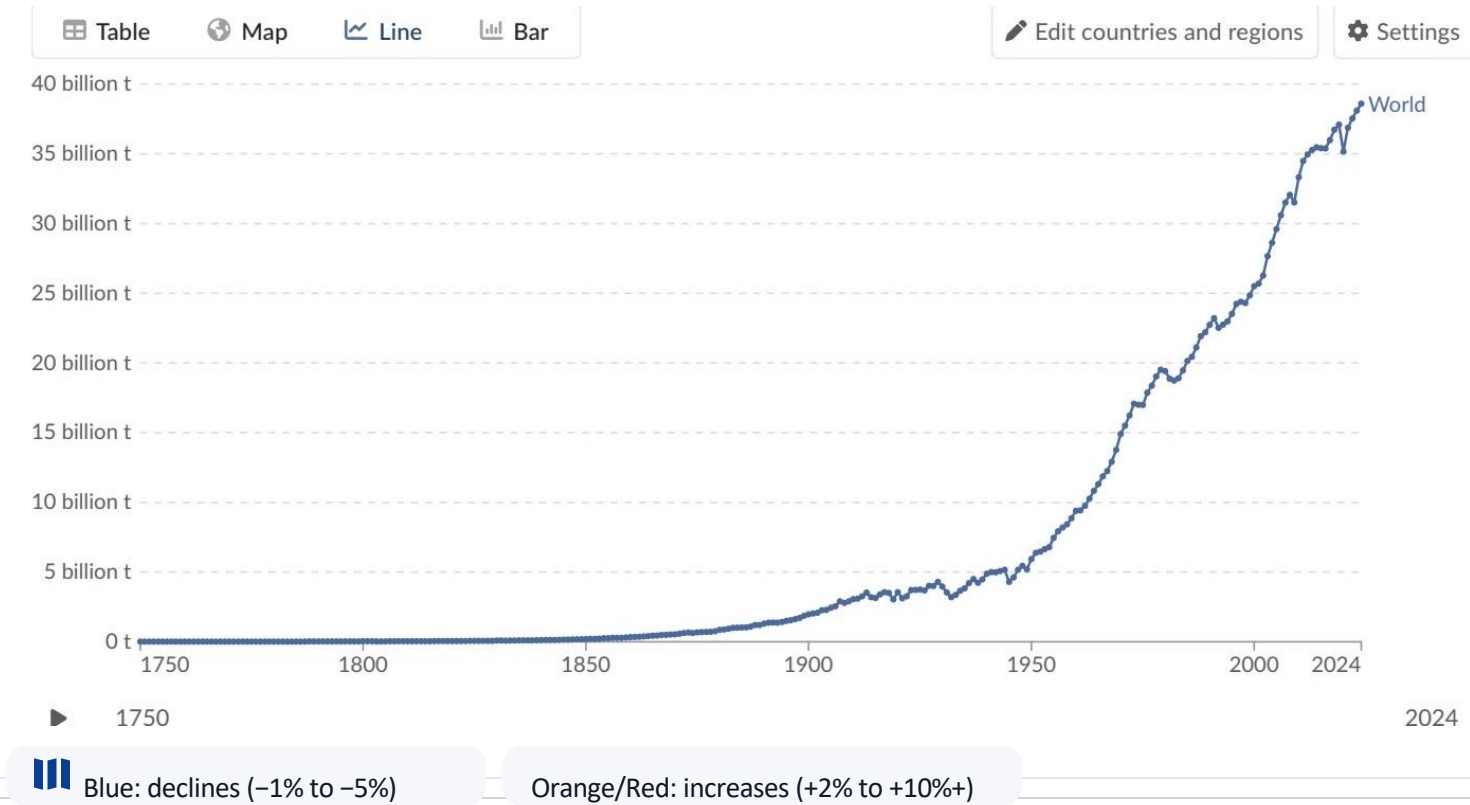


150×
Approx. gap between highest and lowest
“The average American emits in under two days what the average person in Chad/Niger emits in a year.”

2024 shows a mixed picture: US + much of Europe decline (−1% to −5%) while Russia, SE Asia and parts of Africa increase (+2% to +10%+)

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Annual % change in CO₂ emissions, 2024 (figure_4.jpg)



Patterns to note

Declines

- United States
- Much of Europe

Increases

- Russia
- Parts of Southeast Asia
- Several African nations

South America is mixed: some countries show very large increases (>10%), others decline.

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Mechanism

Countries with similar living standards can have different CO₂ outcomes because electricity and heat may be produced from:

- Higher nuclear + renewables (lower CO₂ intensity)
- Higher fossil fuel share (higher CO₂ intensity)

Policy relevance

- Per-capita emissions are not determined by prosperity alone
- Energy system choices can reduce emissions without lowering living standards
- Negotiations can focus on feasible pathways: power sector, industry, efficiency

Illustrative comparison

From provided brief: France/UK lower per-capita than Germany/Netherlands due to energy mix

Country group	Energy source	Per-capita CO ₂
France / UK	Higher nuclear and renewables Lower fossil share	Lower
(Brief cites France and UK ~4 t/person among selected countries)		
Germany / Netherlands	Higher fossil fuel share Lower nuclear (relative)	Higher
(Brief states higher than France/UK; no numeric values provided)		

Note: Table is qualitative to avoid inventing values not in the brief.

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

- ✓ 1) Global emissions exceed 35 billion tonnes/year — not yet peaked
- ✓ 2) China emits ~12.5 Bt (~27%) — more than double the US (~5 Bt)
- ✓ 3) Regional center shifted: Europe+US >90% (1900) → Asia ~50% today; US+Europe < one-third
- ✓ 4) Per-capita inequality is ~150× (US/Australia ~15 t vs Chad/Niger ~0.1 t)
- ✓ 5) Energy mix and policy choices materially shape per-capita outcomes (e.g., France/UK vs Germany/Netherlands)

Implication for negotiations

Responsibility and capability discussions should consider both total emissions (current + historical) and per-capita inequities — while focusing on energy-system levers that change trajectories.